Definition. *If* G *is a graph with vertices* $V(G) = \{v_1, \dots, v_n\}$, *the* ***adjacency matrix*** *of* G *is the* $n \times n$ *matrix*

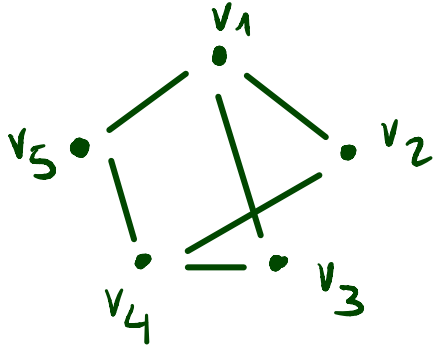
$$A = [a_{ij}], \quad \textit{where} \quad a_{ij} = \begin{cases} 1 & \textit{if } v_i v_j \in E(G), \\ 0 & \textit{otherwise.} \end{cases}$$

Good idea for storing graphs

- The adjacency matrix is symmetric. $a_{ij} = a_{ji}$ $(v_i v_j \in E(G) \iff v_j v_i \in E(G))$.
- The diagonal entries are all 0.
- The number of 1's on the j-th column equals $\deg v_j$

$[\ \#\ ] \to a_{ij} = 1 \iff v_i v_j \in E \iff v_i \in N(v_j)$

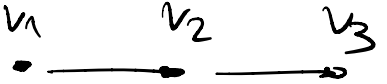
Example. *Compute the adjacency matrix of the following graph*



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Graphs $\longrightarrow$ Matrices

Theorem. *For each positive integer* k*, and each* $i, j \in \{1, \dots, n\}$*, the* (i,j)*-th entry of the matrix* A^k *counts the number of* v_i–v_j *walks of length exactly* k*.*

E.g: P_3 

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$i=1, j=2$: $v_1 - v_2$ walks

$v_1 - v_3$

length 1 $\rightsquigarrow$ 0 $= a_{13}$

length 1 $\rightsquigarrow$ 1 $= a_{12}$

length 2 $\rightsquigarrow$ 1 $= (A^2)_{13}$

length 2 $\rightsquigarrow$ 0 $= (A^2)_{12}$

$$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

Theorem. *For each positive integer* k*, and each* $i, j \in \{1, \dots, n\}$*, the* (i,j)*-th entry of the matrix* A^k *counts the number of* v_i–v_j *walks of length exactly* k.

$(B)_{ij} = (i,j)$ th entry of B

Proof: For $k = 1$

$$\#\text{ of } v_i - v_j \text{ walks of length } 1 = \begin{cases} 1 & \text{if } v_i v_j \in E(G) \\ 0 & \text{otherwise} \end{cases} = a_{ij} = (A^1)_{ij}$$

Assume true for walks of length k. WTS true for length $k+1$.

A $v_i - v_j$ walk of length $k+1$ is just a $v_i - v_t$ walk of length k for some neighbor v_t of v_j, plus the edge $v_t v_j$. Then

$$\#\ v_i - v_j \text{ walks of length } k+1 = \sum_{v_t \in N(v_j)} (A^k)_{it} \cdot 1 = \sum_{t=1}^{n} (A^k)_{it}\, a_{tj}$$

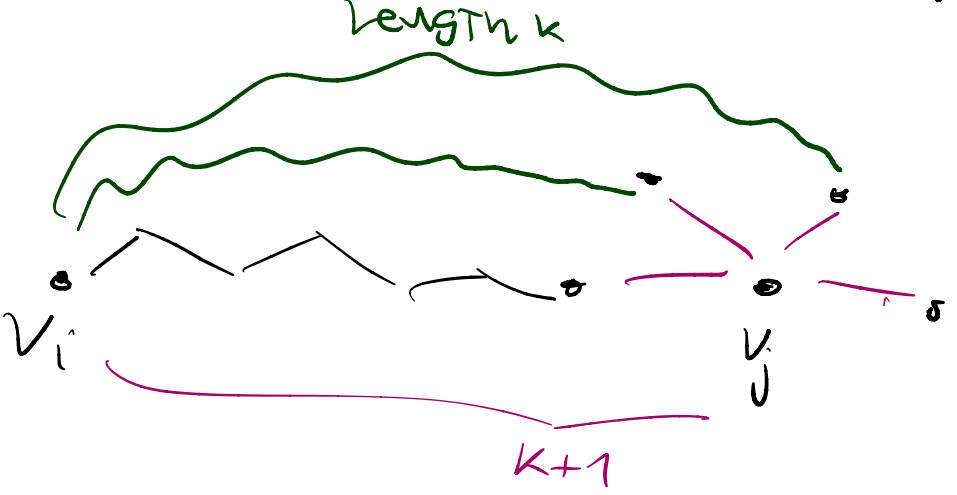
$(A^k)_{it}$: # $v_i - v_t$ walks of length k (H.I.)

a_{tj}: 1 if $v_t \in N(v_j)$, 0 otherwise

$$= \sum_{\tau=1}^{n} (A^k)_{i\tau} (A^1)_{\tau j}$$

$$= (A^k A)_{ij}$$

$$= (A^{k+1})_{ij}$$



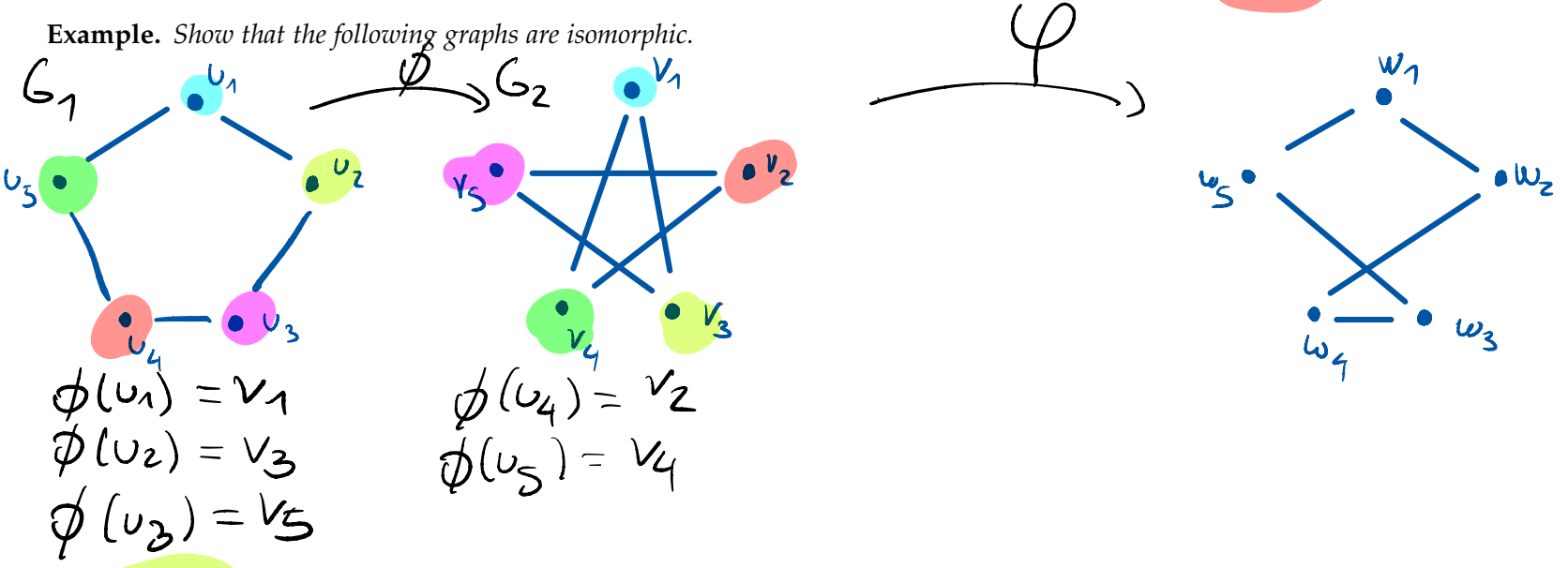
Previously: Two graphs G, H are equal if $V(G) = V(H)$ and $E(G) = E(H)$.

Definition. • *An **isomorphism** from G to H is a bijective function $\phi\colon V(G) \to V(H)$ such that*

$$uv \in E(G) \text{ if and only if } \phi(u)\phi(v) \in E(H).$$

• *If there is an isomorphism ϕ as above, we say that G and H are **isomorphic**, and denote $G \cong H$ or $\phi\colon G \xrightarrow{\cong} H$.*

Example. *Show that the following graphs are isomorphic.*



Yoga: If $G \cong H$, then we can translate any property or construction from G to H (and vice versa).

Theorem. *Assume* $\phi: G \xrightarrow{\cong} H$ *is an isomorphism. Then*

(a) $|V(G)| = |V(H)|$ *and* $|E(G)| = |E(H)|$

(b) $(v_0, v_1, \ldots, v_n)$ *is a* {*path, walk, trail, cycle*} *in* G *iff* $(\phi(v_0), \phi(v_1), \ldots, \phi(v_n))$ *is a* {*path, walk, trail, cycle*} *in* H.

(c) G *is connected iff* H *is connected.*

(d) G *is bipartite iff* H *is bipartite.*

(e) For any $v \in V(G)$, $N_H(\phi(v)) = \{\phi(u) : u \in N_G(u)\}$, *i.e.,* ϕ *induces a bijection from* $N_G(v)$ *to* $N_H(\phi(v))$.

(f) For any $v \in V(G)$, $\deg_G(v) = \deg_H(\phi(v))$. *Thus* G *and* H *have the same degree sequences.*

(g) $\delta(G) = \delta(H)$ *and* $\Delta(G) = \Delta(H)$. *In particular,* G *is* r*-regular iff* H *is* r*-regular.*

Proof (a) Since $\phi: V(G) \longrightarrow V(H)$ is a one-one correspondence, we have $|V(G)| = |V(H)|$. For edges, ϕ also induces a corresp. $uv \in E(G) \iff \phi(u)\phi(v) \in E(H)$, This gives $|E(G)| = |E(H)|$.

(b) $(v_0, v_1, \ldots, v_n)$ is a walk $\iff v_i v_{i+1} \in E(G)$ for all $i \in \{0, \ldots, n-1\}$
$\iff \phi(v_i)\phi(v_{i+1}) \in E(H)$ for all i
$\iff (\phi(v_0), \phi(v_1), \ldots, \phi(v_n))$ is a walk.

$(v_0, \ldots, v_n)$ is a path $\iff$ $(v_0, \ldots, v_n)$ is a walk and $v_i \neq v_j$ if $i \neq j$

$\iff$ $(\phi(v_0), \ldots, \phi(v_n))$ is a walk and $\phi(v_i) \neq \phi(v_j)$ if $i \neq j$

ϕ is one-one

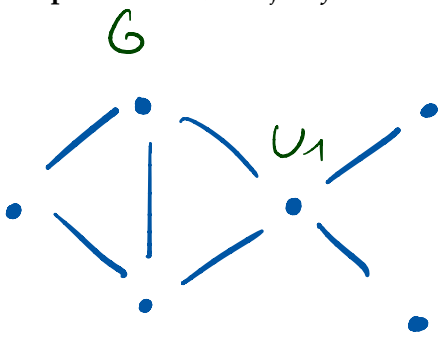
$\iff$ $(\phi(v_0), \ldots, \phi(v_n))$ is a path.

c) Any two vertices in G are connected by a path

$\iff$ any two vertices in H are connected by a path (b).

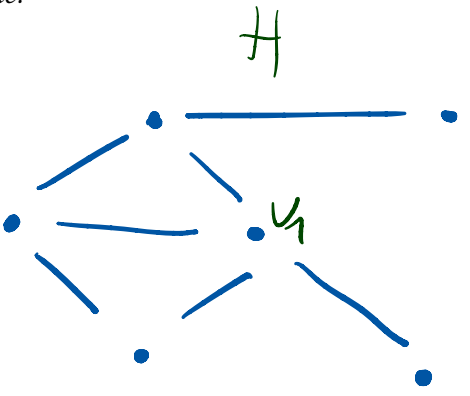
d) Exercise: use cycles + part b.

Example. *Decide (with justification) whether the following graphs are isomorphic.*



4, 3, 3, 2, 1, 1

isomorphic?



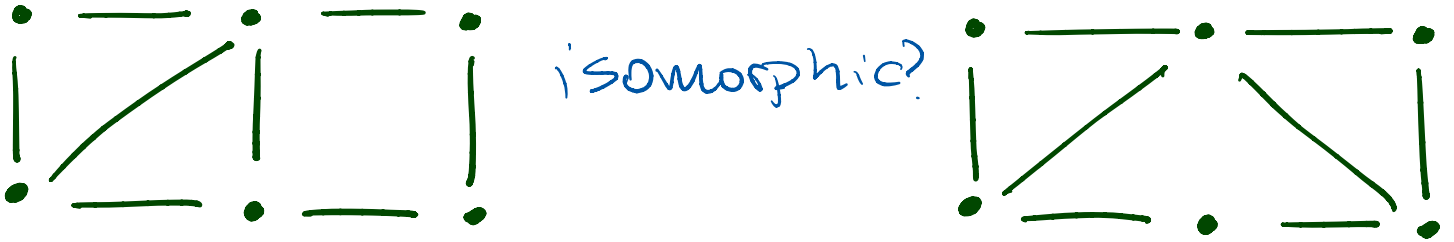
4, 3, 3, 2, 1, 1

Assume $\phi : G \longrightarrow H$ isomorphism.

H has a path of length 5, G not.

If $\exists \phi$, $\phi(u_1) = v_1$. But u_1 has 2 neighbours of deg 1 not the case for v_1, $\not\exists \phi$.

Example. *Decide (with justification) whether the following graphs are isomorphic.*



Recall: A binary relation R on a set X is an **equivalence relation** if it is:

(a) **reflexive:** xRx for all x in X;

(b) **symmetric:** if xRy then yRx for all $x, y \in X$;

(c) **transitive:** if xRy and yRz then xRz for all $x, y, z \in X$.

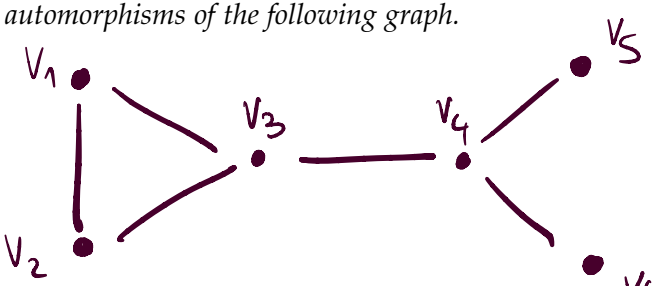
Theorem. *The isomorphism relation* $G R H$ *iff* $G \cong H$ *defines an equivalence relation on the set of all graphs.*

Definition. *An* ***automorphism*** *of a graph* G *is an isomorphism from* G *to itself,* Aut(G) *denotes the set of automorphisms of* G.

- Aut(G) is actually a **group** under composition.
- The operation G ↦ Aut(G) relates graph theory to group theory.

Graph Theory ⟶ Group Theory

Example. *Exhibit automorphisms of the following graph.*



$\alpha(v_1) = v_2$, $\alpha(v_2) = v_1$, and $\alpha(v_i) = v_i$ $i \neq 1, 2$.

$\beta(v_5) = v_6$, $\beta(v_6) = v_5$, $\beta(v_i) = v_i$ $i \neq 5, 6$.

Last one: do both α, β.